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9702/23

October/November 2014

1 hour

No Additional Materials are required.

READ THESE INSTRUCTIONS FIRST

Write in dark blue or black pen.

Do not use staples, paper clips, glue or correction fluid.

DO **NOT** WRITE IN ANY BARCODES.

Answer **all** questions.

Electronic calculators may be used.

You may lose marks if you do not show your working or if you do not use appropriate units.

At the end of the examination, fasten all your work securely together.

The number of marks is given in brackets [] at the end of each question or part question.

This document consists of **14** printed pages and **2** blank pages.

Data

speed of light in free space,	$c = 3.00 \times 10^8 \text{ m s}^{-1}$
permeability of free space,	$\mu_0 = 4\pi \times 10^{-7} \text{ H m}^{-1}$
permittivity of free space,	$\epsilon_0 = 8.85 \times 10^{-12} \text{ F m}^{-1}$
	$(\frac{1}{4\pi\epsilon_0} = 8.99 \times 10^9 \text{ m F}^{-1})$
elementary charge,	$e = 1.60 \times 10^{-19} \text{ C}$
the Planck constant,	$h = 6.63 \times 10^{-34} \text{ J s}$
unified atomic mass constant,	$u = 1.66 \times 10^{-27} \text{ kg}$
rest mass of electron,	$m_e = 9.11 \times 10^{-31} \text{ kg}$
rest mass of proton,	$m_p = 1.67 \times 10^{-27} \text{ kg}$
molar gas constant,	$R = 8.31 \text{ J K}^{-1} \text{ mol}^{-1}$
the Avogadro constant,	$N_A = 6.02 \times 10^{23} \text{ mol}^{-1}$
the Boltzmann constant,	$k = 1.38 \times 10^{-23} \text{ J K}^{-1}$
gravitational constant,	$G = 6.67 \times 10^{-11} \text{ N m}^2 \text{ kg}^{-2}$
acceleration of free fall,	$g = 9.81 \text{ m s}^{-2}$

Formulae

uniformly accelerated motion,

$$s = ut + \frac{1}{2}at^2$$

$$v^2 = u^2 + 2as$$

work done on/by a gas,

$$W = p\Delta V$$

gravitational potential,

$$\phi = -\frac{Gm}{r}$$

hydrostatic pressure,

$$p = \rho gh$$

pressure of an ideal gas,

$$p = \frac{1}{3} \frac{Nm}{V} \langle c^2 \rangle$$

simple harmonic motion,

$$a = -\omega^2 x$$

velocity of particle in s.h.m.,

$$v = v_0 \cos \omega t$$

$$v = \pm \omega \sqrt{(x_0^2 - x^2)}$$

electric potential,

$$V = \frac{Q}{4\pi\epsilon_0 r}$$

capacitors in series,

$$1/C = 1/C_1 + 1/C_2 + \dots$$

capacitors in parallel,

$$C = C_1 + C_2 + \dots$$

energy of charged capacitor,

$$W = \frac{1}{2} QV$$

resistors in series,

$$R = R_1 + R_2 + \dots$$

resistors in parallel,

$$1/R = 1/R_1 + 1/R_2 + \dots$$

alternating current/voltage,

$$x = x_0 \sin \omega t$$

radioactive decay,

$$x = x_0 \exp(-\lambda t)$$

decay constant,

$$\lambda = \frac{0.693}{t_{1/2}}$$

Answer **all** the questions in the spaces provided.

- 1 (a) The kilogram, metre and second are SI base units.

State two other base units.

1.

2.

[2]

- (b) Determine the SI base units of

(i) stress,

SI base units[2]

(ii) the Young modulus.

SI base units[1]

- 2 A microphone detects a musical note of frequency f . The microphone is connected to a cathode-ray oscilloscope (c.r.o.). The signal from the microphone is observed on the c.r.o. as illustrated in Fig. 2.1.

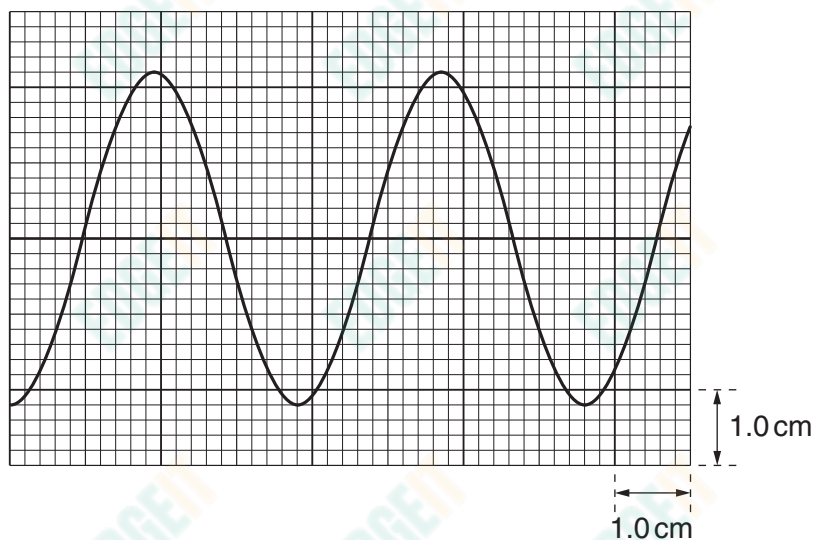


Fig. 2.1

The time-base setting of the c.r.o. is 0.50 ms cm^{-1} . The Y-plate setting is 2.5 mV cm^{-1} .

- (a) Use Fig. 2.1 to determine

- (i) the amplitude of the signal,

amplitude = mV [2]

- (ii) the frequency f ,

f = Hz [3]

- (iii) the actual uncertainty in f caused by reading the scale on the c.r.o.

actual uncertainty = Hz [2]

- (b) State f with its actual uncertainty.

f = $\pm$ Hz [1]

- 3 (a) Force is a vector quantity. State three other vector quantities.

1.
2.
3.

[2]

- (b) Three coplanar forces X , Y and Z act on an object, as shown in Fig. 3.1.

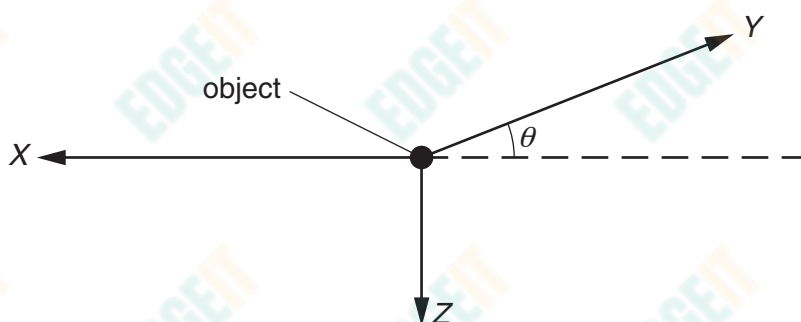


Fig. 3.1

The force Z is vertical and X is horizontal. The force Y is at an angle θ to the horizontal. The force Z is kept constant at 70 N.

In an experiment, the magnitude of force X is varied. The magnitude and direction of force Y are adjusted so that the object remains in equilibrium.

Fig. 3.2 shows the variation of the magnitude of force Y with the magnitude of force X .

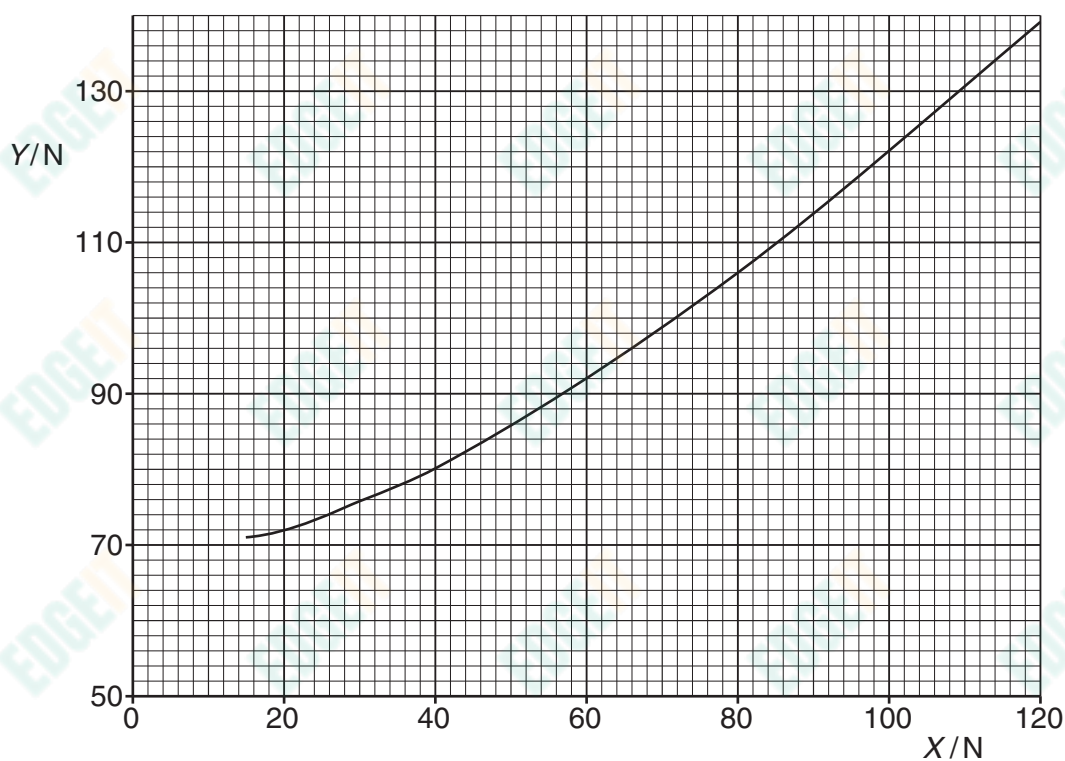


Fig. 3.2

- (i) Use Fig. 3.2 to estimate the magnitude of Y for $X = 0$.

$Y = \dots\dots\dots$ N [1]

- (ii) State and explain the value of θ for $X = 0$.

.....

[2]

- (iii) The magnitude of X is increased to 160 N. Use resolution of forces to calculate the value of

1. angle θ ,

$\theta = \dots\dots\dots^\circ$ [2]

2. the magnitude of force Y .

$Y = \dots\dots\dots$ N [2]

- (c) The angle θ decreases as X increases. Explain why the object cannot be in equilibrium for $\theta = 0$.

.....

[1]

- 4 (a) State the principle of conservation of momentum.

.....

.....

.....[2]

- (b) A ball X and a ball Y are travelling along the same straight line in the same direction, as shown in Fig. 4.1.



Fig. 4.1

Ball X has mass 400 g and horizontal velocity 0.65 ms^{-1} .
 Ball Y has mass 600 g and horizontal velocity 0.45 ms^{-1} .

Ball X catches up and collides with ball Y. After the collision, X has horizontal velocity 0.41 ms^{-1} and Y has horizontal velocity v , as shown in Fig. 4.2.



Fig. 4.2

Calculate

- (i) the total initial momentum of the two balls,

momentum = N s [3]

- (ii) the velocity v ,

$v = \dots \text{ ms}^{-1}$ [2]

- (iii) the total initial kinetic energy of the two balls.

kinetic energy = J [3]

- (c) Explain how you would check whether the collision is elastic.

.....
.....[1]

- (d) Use Newton's third law to explain why, during the collision, the change in momentum of X is equal and opposite to the change in momentum of Y.

.....
.....
.....
.....[2]

- 5 Distinguish between *evaporation* and *boiling*.

evaporation:

.....

.....

boiling:

.....

.....

[4]

- 6 (a) A wire has length 100cm and diameter 0.38mm. The metal of the wire has resistivity $4.5 \times 10^{-7} \Omega \text{m}$.

Show that the resistance of the wire is 4.0Ω .

[3]

- (b) The ends B and D of the wire in (a) are connected to a cell X, as shown in Fig. 6.1.

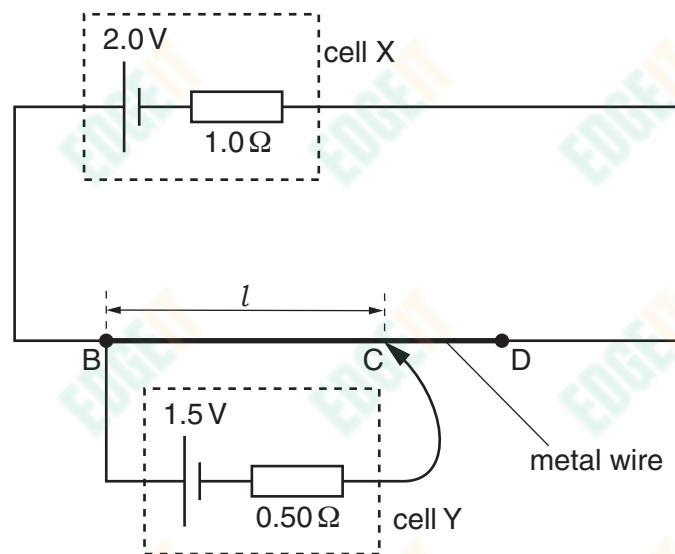


Fig. 6.1

The cell X has electromotive force (e.m.f.) 2.0V and internal resistance 1.0Ω .

A cell Y of e.m.f. 1.5V and internal resistance 0.50Ω is connected to the wire at points B and C, as shown in Fig. 6.1.

The point C is distance l from point B. The current in cell Y is zero.

Calculate

- (i) the current in cell X,

current = A [2]

(ii) the potential difference (p.d.) across the wire BD,

p.d. = V [1]

(iii) the distance l .

l = cm [2]

(c) The connection at C is moved so that l is increased. Explain why the e.m.f. of cell Y is less than its terminal p.d.

.....

[2]

- 7 (a) (i) Explain what is meant by a *progressive transverse wave*.

progressive:

.....

transverse:

.....

[2]

- (ii) Define frequency.

.....

.....[1]

- (b) The variation with distance x of displacement y for a transverse wave is shown in Fig. 7.1.

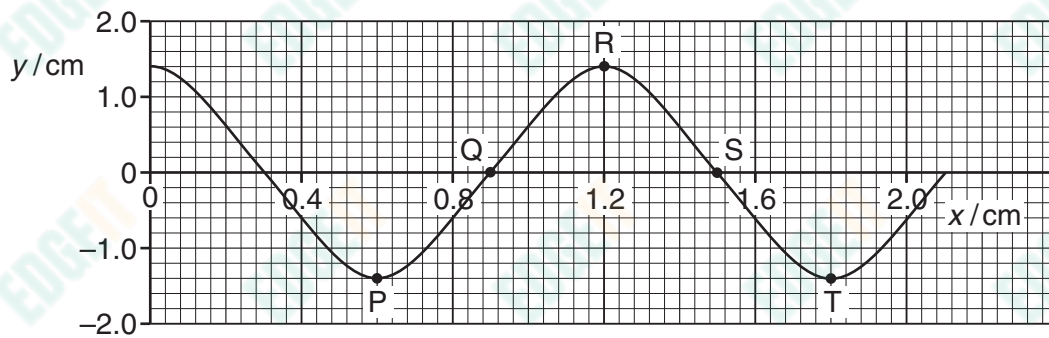


Fig. 7.1

On Fig. 7.1, five points are labelled.

Use Fig. 7.1 to state any two points having a phase difference of

- (i) zero,

.....[1]

- (ii) 270° .

.....[1]

- (c) The frequency of the wave in (b) is 15 Hz.

Calculate the speed of the wave in (b).

speed = ms^{-1} [3]

- (d) Two waves of the same frequency have amplitudes 1.4 cm and 2.1 cm.

Calculate the ratio

$$\frac{\text{intensity of wave of amplitude 1.4 cm}}{\text{intensity of wave of amplitude 2.1 cm}}$$

ratio = [2]

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